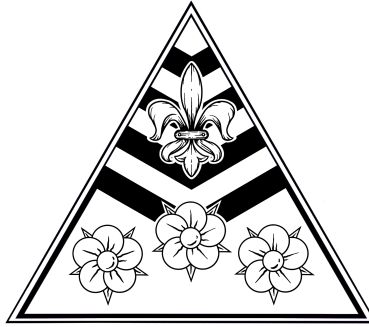
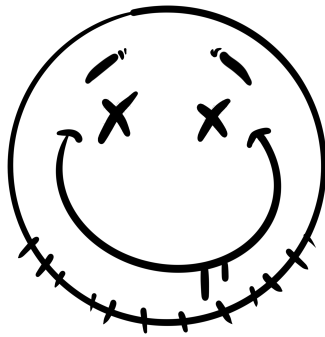




## Lab 12: Sysmon EVTX Forensic Triage

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# 1 Security Event Triage Memorandum

## 1. Engagement Overview

Greater Denver Health Partners requested forensic triage of a suspected security event on an internal Windows host. Evidence was provided as a Windows Event Log export (.evtx) and received by the examining analyst on 2026-05-04. Analysis was conducted per the scenario brief supplied by the client.

## 2. What Was Examined

**Filename:** `psexec-system-execution.evtx` — sourced from the `Microsoft-Windows-Sysmon/Operational` channel on host `fs03vuln.offsec.lan` (offsec.lan domain). SHA-256 hash recorded at intake: `2652560B...874039886` (see Chain of Custody log). Key Event Record IDs examined: 48595, 48596, 48601, and 48605; spanning 2021-12-04T21:17:29Z – 2021-12-04T21:19:16Z.

## 3. Findings

Evidence indicates a threat actor conducted lateral movement onto `fs03vuln.offsec.lan` using PsExec. At 21:17:29Z (EID 48595), `PSEXEC64.exe` — run from `C:\TOOLS\` on an attacker-controlled host — dropped the service binary `PSEXESVC.exe` into `C:\Windows\` on the target via SMB Admin Share. Immediately following (EID 48596), the Windows Service Control Manager wrote the `PSEXESVC` service entry to the registry with a demand-start value (DWORD 0x3), confirming installation. At 21:19:16Z, `PSEXESVC.exe` created a named pipe (EID 48601) and `PSEXEC64.exe` connected to it (EID 48605), completing a remote command execution session. All activity ran under `S-1-5-18` (SYSTEM), indicating the attacker operated with the highest available local privilege. (See IOC Log and ATT&CK Mapping on following pages.)

| IOC                       | Type         | PoP Tier | Notes                                              |
|---------------------------|--------------|----------|----------------------------------------------------|
| <code>\PSEXESVC-F</code>  | Named Pipe   | Annoying | EID 48605 — naming tied to PsExec tool behavior    |
| <code>\PSEXESVC-F</code>  | Named Pipe   | Annoying | EID 48601 — created by PSEXESVC on target host     |
| <code>C:\Windows\</code>  | File Path    | Easy     | EID 48595 — service binary dropped by PsExec64.exe |
| <code>C:\TOOLS\Ps</code>  | File Path    | Easy     | EID 48595/48605 — attacker-side execution binary   |
| <code>HKLM\...\Ser</code> | Registry Key | Annoying | EID 48596 — SCM demand-start entry (DWORD 0x3)     |

## 4. ATT&CK Mapping

| ID        | Technique                          | Tactic                           | Evidence                                                                    |
|-----------|------------------------------------|----------------------------------|-----------------------------------------------------------------------------|
| T1569.002 | System Services: Service Execution | Execution                        | EID 48595 (file drop); EID 48596 (SCM registry write for PSEXESVC)          |
| T1021.002 | Remote SMB/Windows Shares          | Services: Admin Lateral Movement | EID 48601/48605 (named pipes over SMB); RuleName flags Windows Admin Shares |

## 5. Recommendations

IT leadership should hunt across all domain-joined workstations for `PSEXESVC.exe` in `C:\Windows\` and the registry key `HKLM\System\CurrentControlSet\Services\PSEXESVC`, both reliable indicators of PsExec lateral movement. Sysmon EID 17/18 queries filtered for pipe names matching `\PSEXESVC-*` will surface any additional hosts where this technique executed. Analysts should audit non-standard directories (e.g., `C:\TOOLS\`) for PsExec variants and review SMB logs for internal `ADMIN$` share connections. Any newly identified hosts should have their event logs preserved immediately before retention windows expire.

## 2 Chain of Custody Log

| Date<br>/<br>Time<br>(UTC) | Analyst      | Action                           | Evidence Item                       | SHA-256 Before                                                   | SHA-256 After                                                    |
|----------------------------|--------------|----------------------------------|-------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| <b>2026-05-04 22:46 Z</b>  | Keith Schoen | lab12-evidence-bundle_7a166f.zip | Received bundle through lab posting | Not provided                                                     | 615A2D61ACF393F359345B36FC60A822D77B9367E81B2B58852F28478659738F |
| <b>2026-05-04 22:47 Z</b>  | Keith Schoen | MANIFEST.txt                     | Extracted to working directory      | None Provided                                                    | 615A2D61ACF393F359345B36FC60A822D77B9367E81B2B58852F28478659738F |
| <b>2026-05-04 22:47 Z</b>  | Keith Schoen | scenario-brief.txt               | Extracted to working directory      | afa40937c68a8b25a13cddc35a9273b724dd59b14d358289254ae1a92cef56c1 | AFA40937C68A8B25A13CDDC35A9273B724DD59B14D358289254AE1A92CEF56C1 |
| <b>2026-05-04 22:47 Z</b>  | Keith Schoen | psexec-system-execution.evtx     | Extracted to working directory      | 2652560b4da35c983974408316b15a5932c6e69fa21a5193c769622874039886 | 2652560B4DA35C983974408316B15A5932C6E69FA21A5193C769622874039886 |
| <b>2026-05-04 22:48 Z</b>  | Keith Schoen | scenario-brief.txt               | Review documentation                | afa40937c68a8b25a13cddc35a9273b724dd59b14d358289254ae1a92cef56c1 | AFA40937C68A8B25A13CDDC35A9273B724DD59B14D358289254AE1A92CEF56C1 |
| <b>2026-05-04 23:10 Z</b>  | Keith Schoen | psexec-system-execution.evtx     | Opened file in Event Viewer         | 2652560B4DA35C983974408316B15A5932C6E69FA21A5193C769622874039886 | 2652560B4DA35C983974408316B15A5932C6E69FA21A5193C769622874039886 |
| <b>2026-04-05 01:18 Z</b>  | Keith Schoen | psexec-system-execution.evtx     | Review four events in Event Viewer  | 2652560B4DA35C983974408316B15A5932C6E69FA21A5193C769622874039886 | 2652560B4DA35C983974408316B15A5932C6E69FA21A5193C769622874039886 |

++report written with help from Claude (Sonnet 4.6)++